Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Secure File Sharing using Blockchain

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Introduction

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- Growing need for secure file sharing
- Traditional systems are centralized
- problems:
  - Data breaches
  - Single point of failure
  - Lack of transparency

- Solution : Blockchain + IPFS - a secure & decentralized system

Problem Statement

- Centralized storage is vulnerable
- Data integrity is not guaranteed
- High risk of unauthorized access
- Expensive and inefficient storage

Proposed Solution

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- Use **IPFS** for file storage
- Use **Blockchain** for hash storage
- Use **Smart Contracts** for management

✓ Secure
✓ Transparent
✓ Immutable

System Architecture

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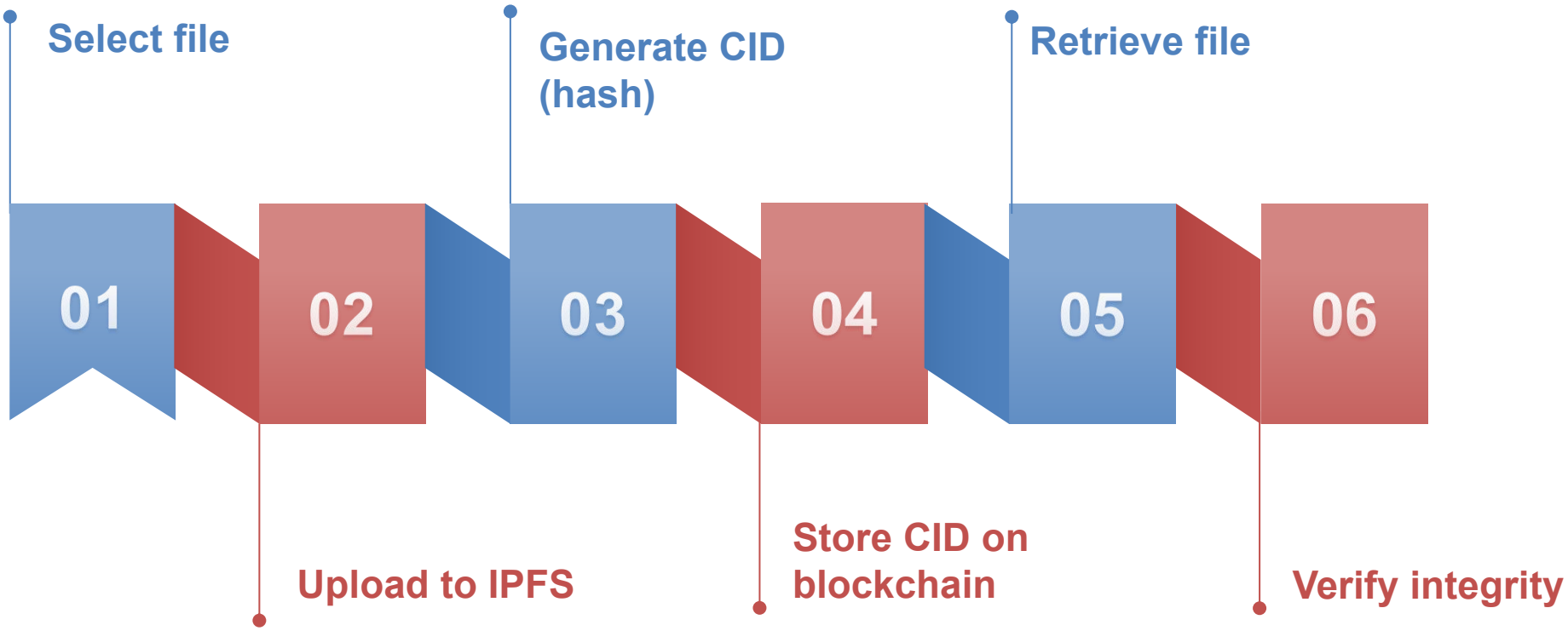
4 Layers:

1. User Interface (React.js)
2. IPFS Storage (Pinata)
3. Blockchain Layer (Ethereum)
4. Security Layer (MetaMask)

Workflow


Select file
Generate CID (hash)
Retrieve file
01
02
03
04
05
06
Upload to IPFS
Store CID on blockchain
Verify integrity

Working Mechanism

- File → IPFS → CID generated
- CID → Stored in blockchain
- Retrieval → Using CID
- Verification → Hash comparison

Tools & Technologies

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- Ethereum (Blockchain)
- IPFS (Storage)
- Pinata (Pinning service)
- Remix IDE (Smart Contracts)
- React.js (Frontend)
- MetaMask (Authentication)

Results

- No single point of failure
- Improved security
- Data integrity ensured
- Transparent System

Advantages

Advantages:

- Secure & immutable
- Decentralized storage
- Data integrity guaranteed
- Transparent system

Limitations:

- Gas fess
- Transaction delay
- IPFS pinning required

Applications

- Healthcare (Patient data)
- Finance (secure documents)
- Education (certificates)
- Supply chain & digital identity

Conclusion

- Successfully built secure system
- Combines blockchain + IPFS
- Ensures integrity and security
- Better than traditional systems

Thank You